

Team Name: RedFox

1. Project Identity

- **Project / Idea Title:** Skyfox
- **Domain:** Robotics, Drones & Smart Automation
- **Idea Describe:** Multi-Purpose Drone Focused For Immediate Response When Delay Of Ambulance and During Mid/Post Disaster Time, Our Project uses an ML dispatch Command to dock payload boxes from a chassis rack for immediate mission execution. It lowers Emergency Healthcare/disaster aid (emergency food, AED/Firstaid kits, and Bluetooth/AI victim detectors). It Also cleans solar panels using dust-density-Detection for air/water sprays, and lands on a ground-station conductive pad to fast-charge and auto-swap payloads for 24/7 operation without manpower Assistance.

2. Concept Overview (Two-Line Summary)

Multi-Purpose Drone Focused For Immediate Response When Delay Of Ambulance and During Mid/Post Disaster Time, Our Project uses an ML dispatch Command to dock payload boxes from a chassis rack for immediate mission execution. It lowers Emergency Healthcare/disaster aid (emergency food, AED/Firstaid kits, and Bluetooth/AI victim detectors). It Also cleans solar panels using dust-density-Detection for air/water sprays, and lands on a ground-station conductive pad to fast-charge and auto-swap payloads for 24/7 operation without manpower Assistance.

3. Problem Statement

- **Disaster Logistics & Safety Hazards:** Manual drone setup wastes critical response time during disaster callouts, while forced ground landings in unstructured debris cause a high rate of propeller strikes and crashes, risking equipment before aid is delivered.
- **Inflexible Single-Purpose Fleet Costs:** Traditional UAVs feature fixed, non-swappable payloads. Purchasing separate single-purpose drone fleets for search-and-rescue, medical drops, and utility maintenance increases overall hardware expenditure by over 60%.
- **Solar Maintenance Efficiency & Resource Waste:** Dust buildup reduces solar panel power output by 15% to 30% monthly. Traditional scheduled water cleaning wastes thousands of liters of clean water on light surface dust that could easily be cleared using dry compressed air.

4. Proposed Solution

Skyfox integrates four core sub-systems into a unified autonomous ecosystem:

- **ML-Driven Dispatch Engine:** Integrates emergency helpline inputs (108) or scheduled maintenance routines to trigger automated selection and mechanical auto-latching of mission-specific payload boxes onto a universal chassis rack.

- **Universal Hover-Winch Assembly:** A high-torque winch tether with a tension-sensing release claw lowers payloads safely from a stationary hover, eliminating propeller crash hazards on uneven rubble or flooded terrain.
- **Modular Mission Payload Suite:**
 - **AED & First-Aid Module:** Delivers emergency medical kits and defibrillators.
 - **Emergency Rations Module:** Drops survival supplies directly to isolated disaster victims.
 - **Victim Detection Module:** Combines ESP32 BLE RSSI signal scanning with onboard AI vision to scan for non-line-of-sight signals from trapped survivors under debris.
 - **Solar Panel Cleaning Modules:** Uses optical dust-density sensors to choose high-pressure **Compressed Air** (for light dust) or targeted **Water Spray** (for heavy grime), achieving up to 80% water savings.
- **Conductive Fast-Charging Base Station:** A self-aligning landing pad with high-current conductive contact plates positioned next to the payload rack recharges the drone and executes automated payload swaps for continuous 24/7 unassisted operation.

5. Innovation & Novelty

- First platform to bridge critical emergency logistics with recurring solar utility maintenance on a single quadrotor chassis.
- Eliminates landing risks through hover-tether winch delivery.
- Reduces solar maintenance water consumption by replacing liquid washing with compressed air based on real-time dust density.
- Enables zero-human-touch operation via continuous fast-charging and automatic payload mechanical latching.

6. Technical Specifications & Feasibility

- **Flight Stack & Electronics:** Pixhawk flight controller with optical precision landing, ROS / ESP32 base-station interface, NEMA stepper winch, high-tensile Kevlar tether, and high-current conductive charging plates.
- **Estimated Prototype Cost:** ₹1.8L – ₹2 L for the Drone, master rack, fast-charging pad, and initial payload box suite.
- **Key Technical Challenges & Mitigation:** Center-of-gravity balancing during fluid dropouts and weatherproofing the exposed conductive landing contacts on the ground dock.

Problem Statement Under SIH 2026 Framework:

Autonomous Mission-Adaptive UAV System for Emergency Disaster Logistics & Infrastructure Maintenance

Price Breakdown For the Required Components to Build

Subsystem	Exact Component & Model	Benchmark Vendor	Unit Retail Price (INR, incl. GST)
Airframe	Tarot 680Pro Hexacopter Foldable Frame (TL68P00)	Robu / IndiaMART	₹13,500 – ₹14,500
Motors	6× Sunnysky V3508 380KV Brushless BLDC Motors	Robu.in	₹17,000 – ₹19,000 (~₹2,900 – ₹3,200/unit)
ESCs	6× 40A OPTO BLHeli/SimonK ESCs	Robu / Robocraze	₹4,800 – ₹5,500 (~₹800 – ₹900/unit)
Propellers	3× Pairs (6 pcs) 1355 / 1555 Carbon-Fiber Props	Robu.in	₹2,400 – ₹2,800
Battery Pack	2× 6S 6000mAh–8000mAh 22.2V LiPo (Orange / Gens Ace)	Robu.in	₹9,500 – ₹11,000

Autopilot	Holybro Pixhawk 6C (Aluminium Case + PM02 Power Module)	Robu.in	₹32,000 – ₹36,000
GNSS GPS	Holybro M9N / Here3+ Precision GPS Compass	Robu / Robocraze	₹5,500 – ₹6,800
Edge AI Board	NVIDIA Jetson Orin Nano Developer Kit (8GB)	Robocraze / Robu	₹24,000 – ₹26,500
SSD Storage	256GB NVMe M.2 SSD + Custom Heatspreader	Robocraze	₹2,200 – ₹2,600
Thermal Camera	FLIR Lepton 3.5 Radiometric Core + PureThermal Breakout	Robu / Mouser India	₹26,000 – ₹28,500
RGB Camera	Sony IMX477 12.3MP HQ Camera (MIPI-CSI)	Robocraze	₹3,800 – ₹4,200
Proximity Sonar	MaxBotix MB1003 / MB1242 I2CXL-MaxSonar Rangefinder	Robu.in	₹3,800 – ₹4,200
Micro-LiDAR	Benewake TF-Mini Plus LiDAR Sensor (IP65, 12m)	Robu.in	₹4,200 – ₹4,600

Winch Actuation	NEMA 17 Geared Planetary Stepper Motor + TMC2209 Driver	Robu.in	₹2,200 – ₹2,600
Tether Release	12V Open-Frame Pull Solenoid + HX711 Load Cell Interface	Robu / FlyRobo	₹1,400 – ₹1,800
Victim Beacon	ESP32-WROOM-32U NodeMCU + External 2.4GHz Antenna	Robu / FlyRobo	₹550 – ₹650
Solar Sprayer	12V Mini Diaphragm Micro-Pump (4L/min) + Mist Nozzles	Robu.in	₹1,200 – ₹1,500
Base Station Relays	40A DC-DC Solid-State Relay (SSR) + ESP32 Controller	Robocraze	₹1,200 – ₹1,500
Docking Supply	24V 15A 360W SMPS Fast-Charging Power Supply	Robu / Generic	₹2,800 – ₹3,200
Hardware & Rigging	Copper Skids, Kevlar Line, PETG Filament, Standoffs	Robu / Local	₹5,000 – ₹6,000
Medical / Aid Boxes* (not Included in	AED Training Unit + Trauma Pack + Ration Containers	First-aid Supplier	₹60,000 – 80,000

Combined Cost)			
Grand Total (Approx.)	Complete Real-Life PoC Build	Robu / Robocraze	₹1,79,500 – ₹1,99,400

Team:

Arvind Naraynan M R (Team Lead)

Govarthan M (Payload Design & Website Head)

ShriNivas (Infield Communication & Speech Head)

M Dharnesh Bala (Impact Analysis & Presentation Head)